Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/05/2020
PI Name:	Cromer, Walter Ph.D.		
Project Title:	Use of Dietary Histidine to Counteract Lymphatic and Mucosal Immune Dysfunction Due to Space Dysbiosis		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	06/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 6/9/21) NOTE: End date changed to 2/28/2021 per NSSC information (Ed., 8/10/2020) NOTE: End date changed to 8/31/2020 per NSSC information (Ed., 8/3/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alaniz, Robert Ph.D. (Texas A&M University System)		
Grant/Contract No.:	80NSSC19K0438		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of our project was to determine what effect the space flight microbiome had on host immunity and physiological function. There were several objectives associated with this goal. The first was to determine if there were any major metabolomic changes in the gastrointestinal (GI) space flight microbiome and if there were significant shifts in the populations of the samples we were using. Our second objective was to determine if there were major genetic alterations in the immune/lymphatic systems from the same animals. The third objective was to determine if fecal microbiome transplant can recapitulate space flight changes we have observed in the immune and lymphatic systems. Our final objective was to determine if histidine supplementation could reverse any changes seen in fecal transplant. Significance: Determining the impact of the space flight microbiome on intestinal physiology and immunology is critical to understanding future risks that may impact human space exploration. Additionally development of a countermeasure to mitigate these issues is also necessary. This proposal addresses both of these critical issues utilizing a technique that has not yet been employed in the field of space biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Determining the impact of the space flight microbiome/metabolome on the physiology and immunology of animals that have never been to space may seem to only have applications to space exploration but that is incorrect. The space flight environment is an extreme environment with a combination of microgravity and radiation and we (the scientific community) has very little information about the impact of these sorts of environments on both the microbiome and if those changes are clinically relevant. Therefore our studies have a crucial role in determining whether or not we should focus on therapeutic interventions focused on the microbiome of persons exposed to extreme environments (military, natural, and manmade disasters). This study will provide the basic foundation for making this determination.
Task Progress:	To date we have accomplished a majority of the first and second objectives with some delays in the metabolomic analysis. We found that animals flown in both RR3 and RR9 missions had very little change in the species diversity and distribution of the GI microbiome between flight and habitat controls. This is at direct odds to the findings of RR1 on which we had based our hypothesis. However, the basal changes in histidine in RR9 suggests that even though there were very little differences in the microbiome the metabolome may still be altered. It should be noted there were major differences between RR3 and RR9 in terms of both diversity and distribution of the microbiome and interestingly the greatest species level difference was between flight samples of RR9 and RR3. These differences between the studies was expected to be due to differences in vendor, strain, etc. but the fact that the largest differences were found in the flight samples suggests that the space flight environment may impact changes in the microbiome based on genetic background, though this is speculative until more studies are performed. While we have not been able to perform the classical metabolomic analysis of the samples yet due to issues leading to shut down of several laboratories associated with processing and analysis of the samples, we do have the genomic data from the samples (deep RNA sequencing) which we are currently analyzing to determine if there are changes in the genetic pathways associated with key metabolites. We have also performed deep RNA sequencing of tissues of the lymphatic system and we have found that there are numerous changes that have occurred in both the immunological and physiological systems associated with those tissues. In both broad untargeted analysis and more focused gene panel assessment we found striking changes. Among the strongest hits that we found in untargeted analysis was calcium handling (clustered with contractile dysfunction on the sponge chart) and immune dysfunction. Focused analysis showed strong indications of endothelial dysfunction in the lymphatic system along with slight indications of a tissue environment that would favor the development of lymphoma. We have not been able to finish the final objectives of the animal experiments in a majority part due to the shutdown of our animal facilities due to COVID-19 in early March. We also received the final portion of the grant payment on July 24th which would cover the analysis of the tissues generated by the animal studies.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Wang W, Gasheva O, Narayanan A, Zawieja S, Castorena J, Gashev A, Bagher P, Mao V, Willey J, Delp M, Cromer W, Zawieja D. "Analysis of the structure of mesenteric lymph nodes from space flown rodents." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. HRP abstract 20310. , Jan-2020
Abstracts for Journals and Proceedings	Alaniz R, Cromer WE. "Metabolomic changes of the gi tract of space flown rodents." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstract 20302. , Jan-2020
Abstracts for Journals and Proceedings	Wang W, Cromer W, Sanchez E, Suntravat M, Si H, Zhang SL, Yin S, Davies P, Zawieja D. "Effect of Crotalus oreganus Derived Crotamine on Lymphatic Endothelial Cells and Lymph Transport." Experimental Biology 2020, San Diego, CA, April 4-7, 2020. Abstracts. Experimental Biology 2020, San Diego, CA, April 4-7, 2020. , Apr-2020